```
!pip install chromadb sentence-transformers langchain -q
```

```
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ERROR: pip's dependency resolver does not currently take into account all the packages that are in
google-adk 1.28.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.40.
google-adk 1.28.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.40.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have c
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have ope
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you ha
```

```
import chromadb
from sentence_transformers import SentenceTransformer
```

```
# Create client
client = chromadb.Client()

# Create collection (like a table)
collection = client.create_collection(name="rag_collection")
```

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingfac
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datas
  warnings.warn(
modules.json: 100%  349/349 [00:00<00:00, 10.5kB/s]

config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 4.47kB/s]

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable h
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hu
README.md: 10.5k/? [00:00<00:00, 560kB/s]

sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 2.04kB/s]

config.json: 100%  612/612 [00:00<00:00, 25.2kB/s]

model.safetensors: 100%  90.9M/90.9M [00:01<00:00, 208MB/s]

Loading weights: 100%  103/103 [00:00<00:00, 354.54it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expec
tokenizer_config.json: 100%  350/350 [00:00<00:00, 12.9kB/s]

vocab.txt: 232k/? [00:00<00:00, 7.88MB/s]

tokenizer.json: 466k/? [00:00<00:00, 21.7MB/s]

special_tokens_map.json: 100%  112/112 [00:00<00:00, 9.72kB/s]

config.json: 100%  190/190 [00:00<00:00, 7.82kB/s]
```

```
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI that learns from data.",
    "Deep Learning uses neural networks with many layers.",
    "Natural Language Processing deals with text and language data.",
    "RAG combines retrieval and generation for better answers.",
    "Supervised learning uses labeled data.",
    "Unsupervised learning finds hidden patterns.",
    "Reinforcement learning learns using rewards.",
    "Neural networks are inspired by the human brain.",
    "Embeddings convert text into numerical vectors.",
    "Vector databases store embeddings efficiently.",
    "Cosine similarity measures similarity between vectors.",
    "Transformers are used in modern NLP models.",
    "BERT is a popular NLP model.",
    "GPT models generate human-like text.",
    "Semantic search understands meaning not just words.",
    "Tokenization splits text into smaller parts.",
    "Overfitting happens when model memorizes data.",
    "Underfitting happens when model fails to learn.",
    "AI is used in healthcare, finance, and education."
]
```

```
ids = [f"doc{i}" for i in range(1, len(documents)+1)]
```

```
embeddings = model.encode(documents)
```

```
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)
```

```
query = "What is Artificial Intelligence?"

query_embedding = model.encode([query])

results = collection.query(
    query_embeddings=query_embedding,
    n_results=3
)

print(results)
```

```
{'ids': [['doc1', 'doc20', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intelligence i
```

```
print("\n🔍 Retrieved Documents:\n")

for doc in results['documents'][0]:
    print("-", doc)
```

```
🔍 Retrieved Documents:

- Artificial Intelligence is the simulation of human intelligence.
- AI is used in healthcare, finance, and education.
- Machine Learning is a subset of AI that learns from data.
```

```
def retrieve(query):
    query_embedding = model.encode([query])
    results = collection.query(query_embeddings=query_embedding, n_results=2)
    return results['documents'][0]

# Test
print("\n📌 Custom Query Result:\n")
print(retrieve("Explain machine learning"))
```

```
📌 Custom Query Result:

['Machine Learning is a subset of AI that learns from data.', 'Supervised learning uses labeled da
```

```
queries = [
    "What is deep learning?",
    "Explain vector database",
    "What is NLP?"
]

for q in queries:
    print(f"\n🔍 Query: {q}")
```

```
print(f"\n🔎 Query: {q}")
print(retrieve(q))
```

```
🔎 Query: What is deep learning?
['Deep Learning uses neural networks with many layers.', 'Machine Learning is a subset of AI that
```